

LIGHTWEIGHT AI FOR TRYPANOSOMA DETECTION IN MICROSCOPY IMAGES

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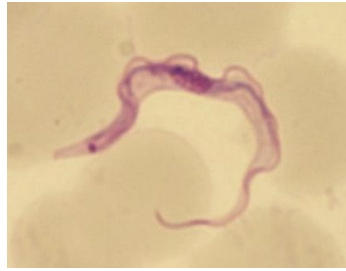
Sys Lab, Dr. Yilmaz, Pd. 7

PROBLEM

- Trypanosoma parasites are extremely dangerous, as delayed diagnosis can lead to severe neurological damage or death. [6]
- Diagnosis relies on manual microscope inspection by trained specialists.
- This process is:
 - Time-consuming (each slide can take several minutes to inspect)
 - Error-prone
 - Limited by resource availability (few labs have skilled microscopists)
- Diagnosis is critical because the disease is fatal if missed



BACKGROUND



What is *Trypanosoma*?

- *Trypanosoma* is a microscopic, single-celled protozoan parasite transmitted by insects.
 - *T. brucei* → causes African Sleeping Sickness (via the tsetse fly)
 - *T. cruzi* → causes Chagas Disease (via the kissing bug)
- Over 70 million people live in regions at risk. [3]
- Without early detection and treatment, infections can cause neurological damage, heart failure, and death.

How Detection Works Today:

- Blood smears are stained and examined under a microscope.
- Diagnosis depends on visually spotting the slender, purple-stained Trypanosome among thousands of red blood cells.

BACKGROUND CONT.

- Artificial intelligence has started to change how we approach medical diagnostics
- Digital microscopy + Artificial Intelligence = automated image analysis
- Deep learning can identify cells, parasites, and patterns that are invisible to the human eye.

Applications in Healthcare:

- Malaria parasite detection
- Tuberculosis classification
- Cancer cell recognition
- Retinal disease screening

Why It Matters:

- **Faster:** AI can process thousands of images in seconds.
- **More consistent:** Reduces human variability.
- **Scalable:** Can assist clinics lacking trained technicians.

OTHER SOLUTIONS

Manual microscopy:

- Gold standard but slow, error-prone, and requires expert skill.

Molecular Diagnostics (PCR, ELISA)

- Detect parasite DNA or antigens
- Highly accurate but expensive and lab-bound

AI models for Trypanosoma:

- Hybrid deep learning model [2]
 - a. Combines object detection and classification.
 - b. Detects *T. cruzi*, *T. brucei*, and *T. evansi* in microscopic images.
- High accuracy on lab-grade images. (~90%)
- Heavy computational requirements (GPU or high-end hardware needed).
- Not optimized for field deployment or mobile use.

NOVELTY

Lightweight / Field-Deployable:

- Optimized for less powerful hardware.
- Can run real-time inference without high-end GPUs.
- 12 MB model, runs on standard laptop CPU

Trypanosoma-Specific AI:

- Focused on Trypanosoma detection, not just general blood pathogens.
- Trained specifically on the Tryp dataset.

Comparison to Existing Models:

- While previous research has demonstrated that deep learning models can detect Trypanosoma effectively, these models are computationally heavy and confined to laboratory environments.

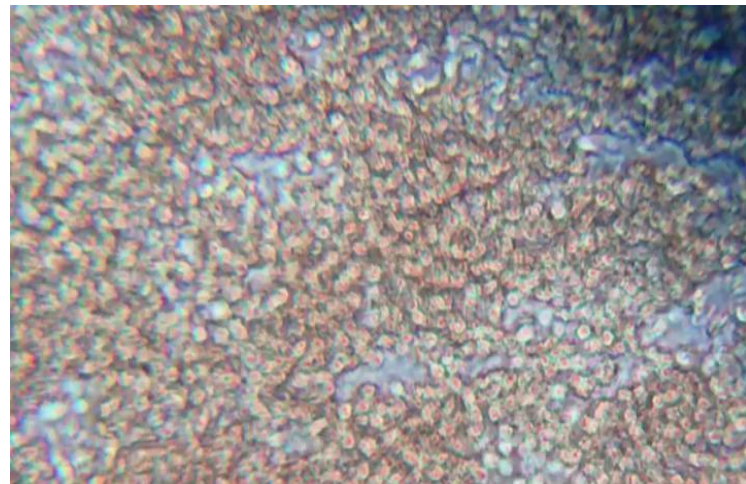
IMPACT

Public Health Relevance:

- Trypanosomiasis is a neglected tropical disease that affects rural and resource-limited regions.
- Early detection can prevent disease progression and fatalities.
- Around 50,000 deaths each year [5]
- Around 100% mortality rate for African sleeping sickness if left untreated
- While existing AI solutions exist, they require GPU hardware unavailable in endemic regions.

DATASET FROM ANZAKU ET AL. [1]

- Microscopy images labeled as Trypanosoma positive/negative (3115 positive, 93 negative)
- Labels contained in JSON file in MS COCO format, including:
 - A list containing image file information (id, file name, width, height)
 - A list of object data (parasite id, corresponding image id, bounding box, area)
 - A list defining the labels (just parasite in this case)



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[{"height": 1024, "width": 1360, "id": 1, "file_name": "positive_video_066_00000089.jpg"},
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METHOD: PREPROCESSING

- Format labels from JSON file to YOLO format
- Train test validation split of 3:1:1

≡ [positive_video_002_00000065.txt](#) ×

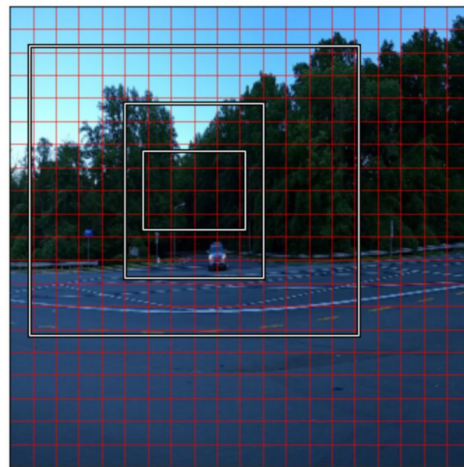
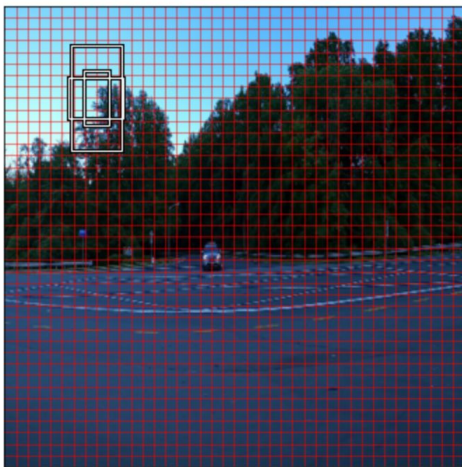
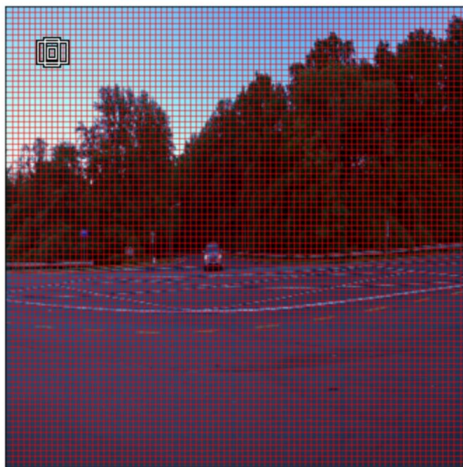
preprocessed > labels > test > ≡ positive_video_002_00000065.txt

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2 0 0.786397 0.922363 0.041912 0.047852
3 0 0.361029 0.946777 0.048529 0.047852
4 0 0.314338 0.210938 0.036029 0.089844
5 0 0.722426 0.455566 0.061029 0.053711
6 0 0.376838 0.610352 0.028676 0.074219
7
```

	# of images	Positive	Negative
Train	1,950	1,893	57
Validation	628	610	18
Test	630	612	18

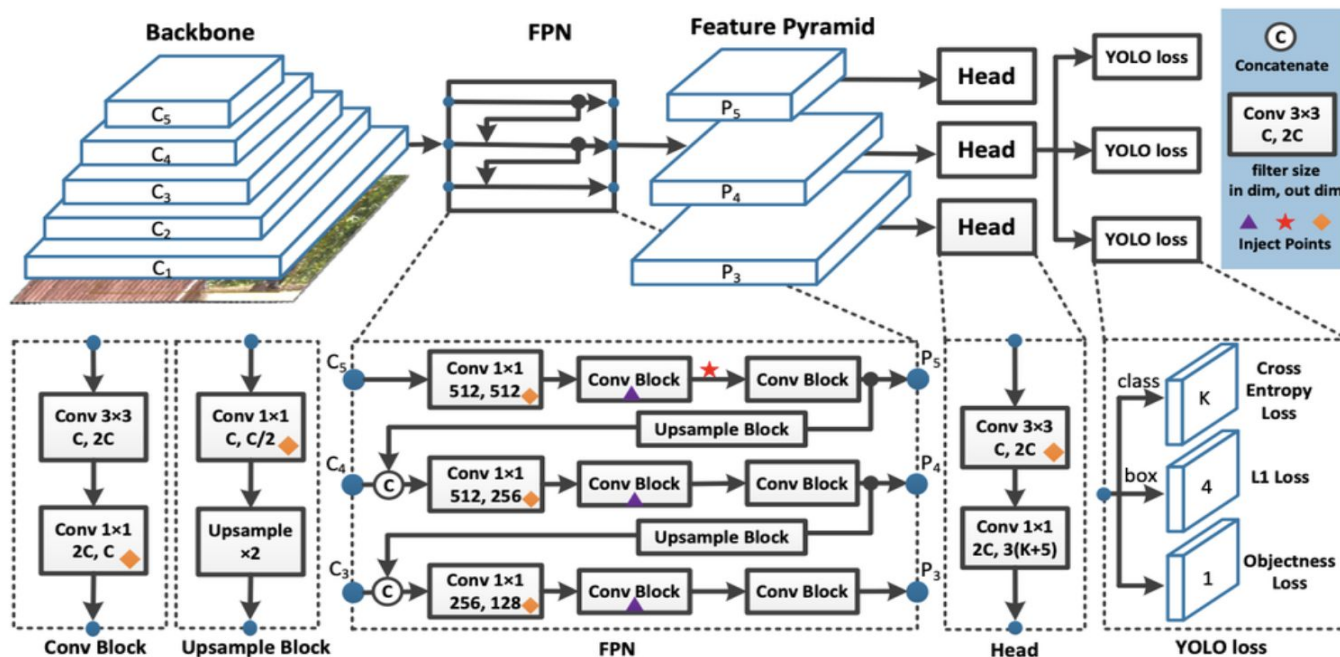
METHOD: HOW DETECTION WORKS

- We don't need negative images because of grid overlay
- Each cell predicts whether it contains a parasite or not. So in a single image with say 5 parasites, there are 5 positive regions and thousands of negative regions



- means >99% of predictions are on negative regions
- this is why precision, recall, and mAP metrics are so important

METHOD: YOLO V7-TINY



In YOLOv7-tiny, these blocks are smaller and use fewer channels than the full model, which is what keeps the parameter count down to about 6 million.

METHOD

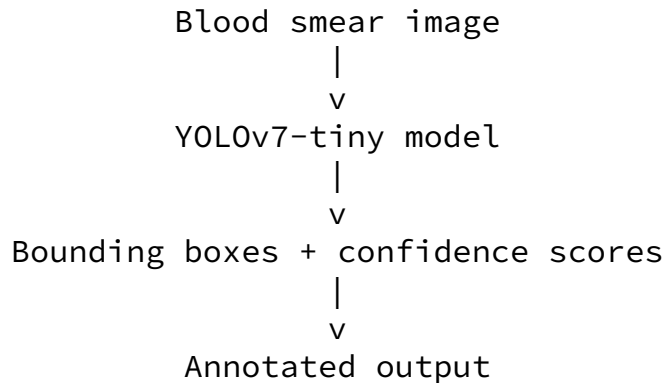
Baseline Training:

- YOLOv7 tiny
- Batch size: 16
- Image size: 640x640
- Epochs: 100
- Optimizer: SGD (lr=0.01, momentum=0.937)
- COCO pretrained base weights
- Hardware: Google Colab T4 GPU (free tier)

Evaluation Metrics:

- Precision – of all detections, how many were correct
- Recall – of all actual parasites, how many were found
- mAP@0.5 – average precision at 50% IoU (primary metric)
- mAP@0.5:0.95 – stricter, averaged across multiple IoU thresholds

Overall Flow:



RESULTS

Precision:

74.9%

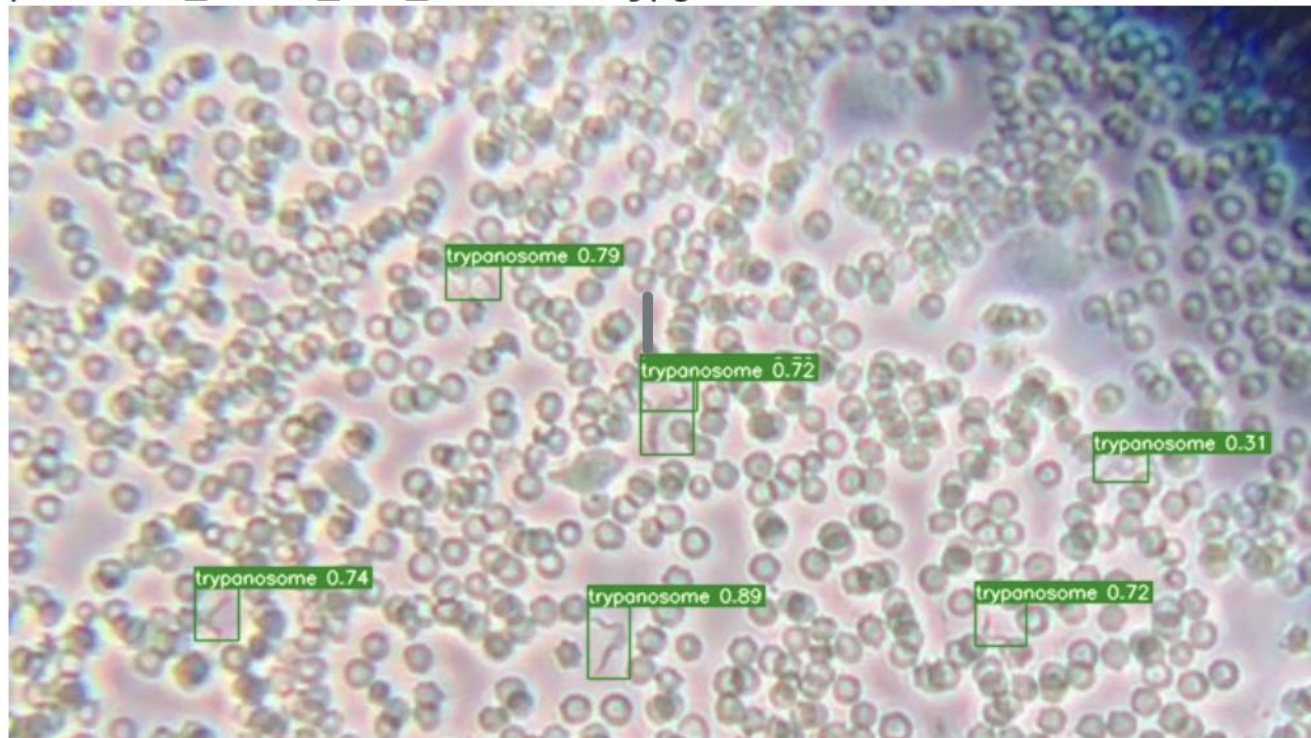
Recall: 70.2%

AP@0.5: 74.1%

AP@0.5:0.95:

34.4%

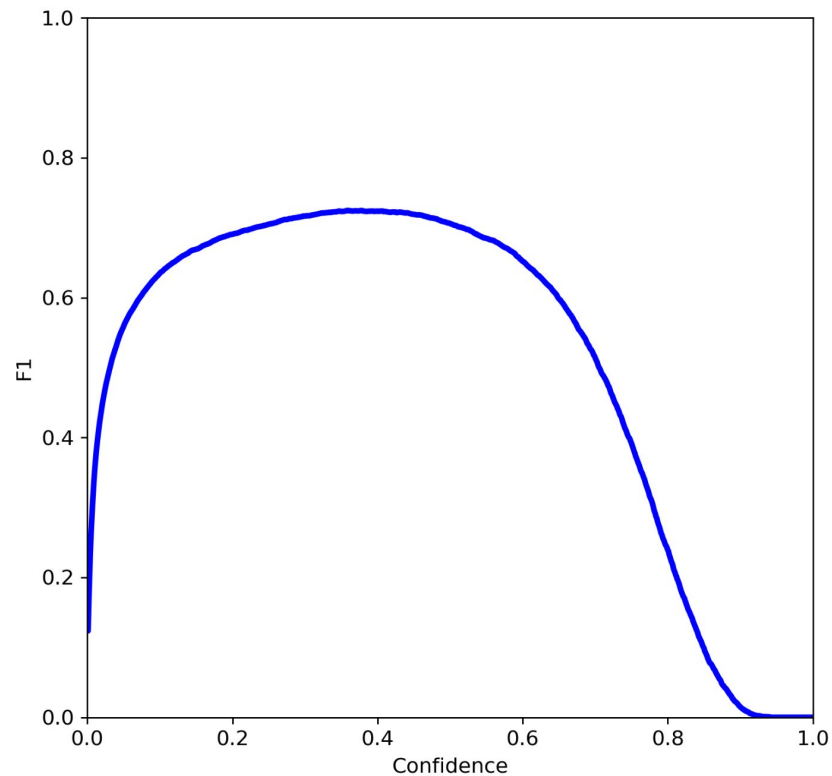
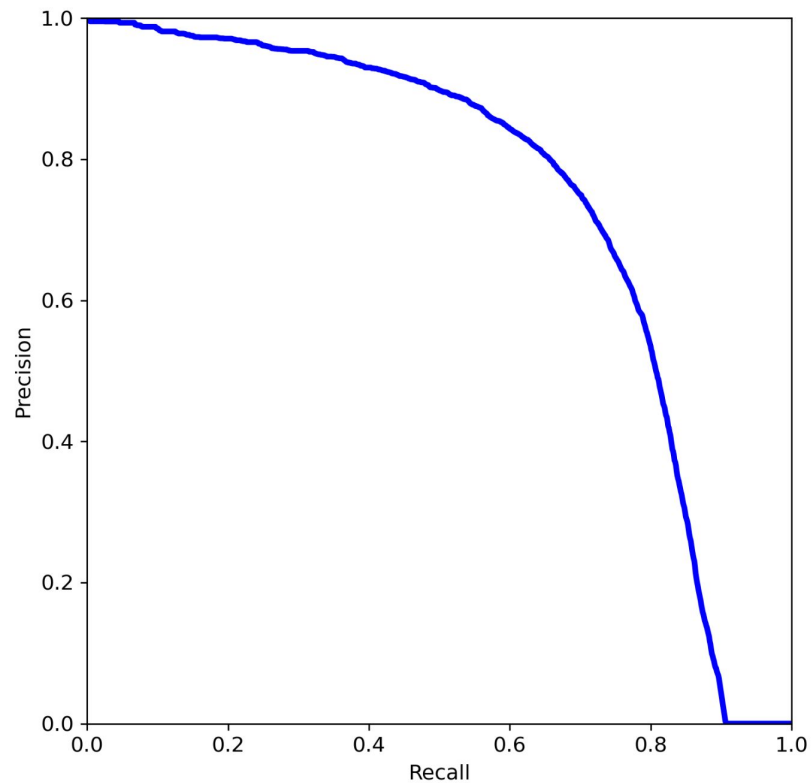
positive_video_023_00000377.jpg



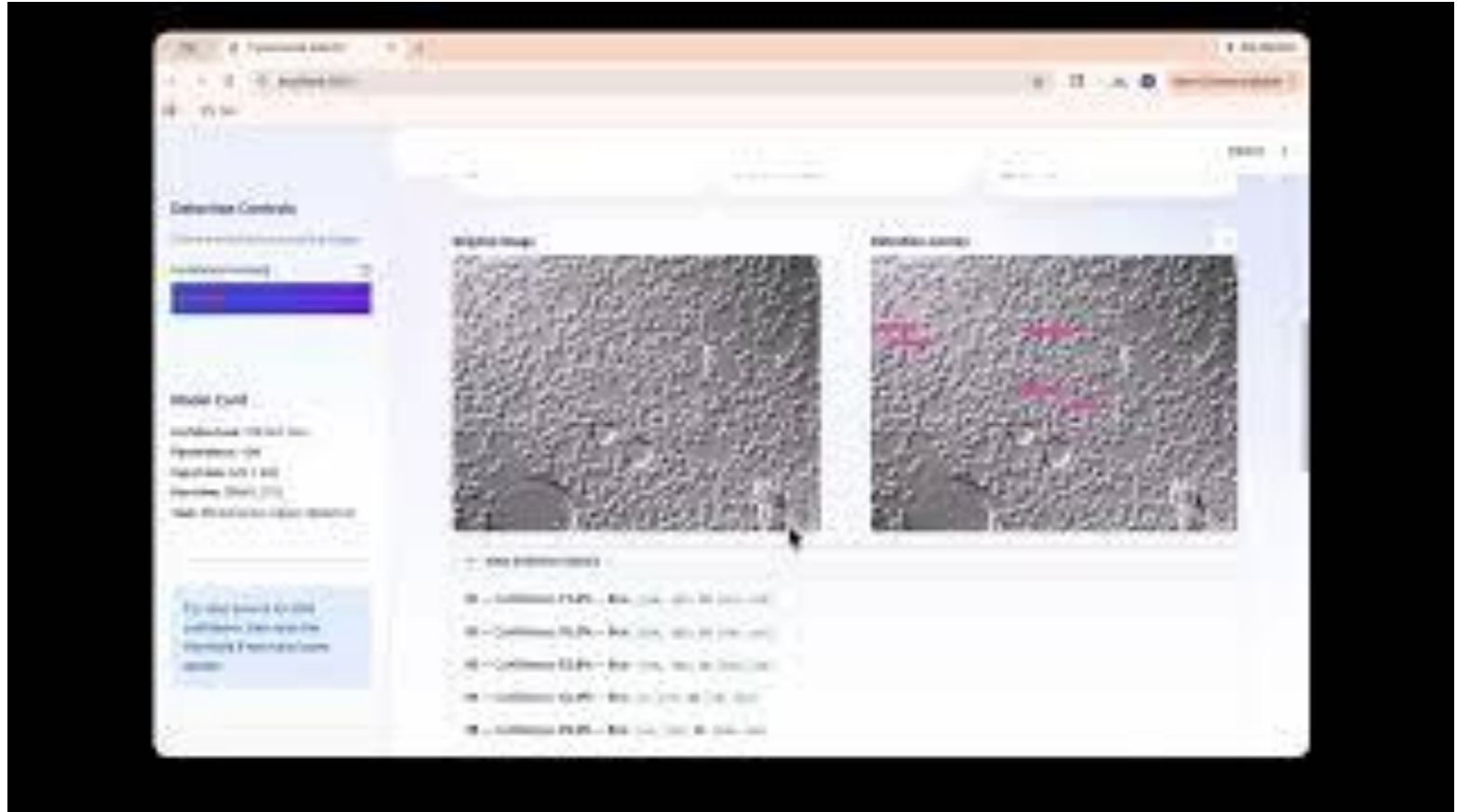
RESULTS CONT.



RESULTS CONT.



DEMO



CONCLUSION AND FUTURE WORK

What we showed

- YOLOv7-tiny can detect Trypanosoma parasites with 74.1% AP@0.5 – competitive with much larger models
- The full pipeline (12 MB model) runs on a standard laptop CPU with no GPU required
- Demonstrates that effective parasite screening is feasible on accessible hardware

Limitations

- 70.2% recall means roughly 30% of parasites are missed – suitable as a screening tool to flag potentially positive samples for human confirmation, not as a standalone diagnostic

Future Work

- Multi-species classification (T. cruzi vs. T. brucei)
- Real-time inference from live microscopy feeds
- Mobile app deployment for field clinicians

REFERENCES

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- [5] Centers for Disease Control and Prevention, "DPDx – Trypanosomiasis, American." [Online]. Available: <https://www.cdc.gov/dpdx/trypanosomiasisamerican/index.html>
- [6] Eisai Co., Ltd., "Diseases of Africa." [Online]. Available: <https://www.eisai.com/sustainability/people/globalhealth/diseases/africa.html>

Q&A?

THANKS!